

A FEW ELEMENTARY FACTS FROM LOCAL BIFURCATION THEORY

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1. GENERIC VECTOR FIELDS

Let us consider a first order autonomous differential equation,

$$(1.1) \quad \dot{x} = V(x)$$

where $x \in \mathcal{C}^1(\mathbb{R}, \mathbb{R}^d)$ and $V \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$. We are interested in the *typical local behaviour* of such systems. Unfortunately, before being able to even address such an issue, it is necessary to give a technical meaning to the three words *typical*, *local* and *behaviour*. By *local* understanding in a region K we mean that for each point $x \in K$ we are able to consider some neighborhood of x in which we understand the solutions of (1.1).¹ To do so it would be nice to consider only neighborhoods U in which $V(x) \neq 0$ with, at most, the exception of one point. Of course, this is not always possible (think of the case $V \equiv 0$), our claim is that *typically* it is.

To define *typical* let us consider

Definition 1. *Given a topological space Ω , we say that a set $A \subset \Omega$ is generic if it is open and dense.*

Now $\mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ is a Banach space² hence the topology is trivially determined by the norm. For now on *typical* will mean that it happens for a countable intersection of *generic* sets (this is also called a *residual* set).

Problem 1. *Prove that a residual set is dense.*

Problem 2. *Prove that, for each compact set $K \subset \mathbb{R}^d$,*

$$A_K := \{V \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d) : V(x) = 0 \text{ implies } D_x V \text{ hyperbolic } \forall x \in K\}$$

is generic. (Hint: Without loss of generality assume $V(0) = 0 \in K$ and $D_0 V = A$ hyperbolic. Then for each vector field $W \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ consider the family $V(x, \mu) := V(x) + \mu W(x)$. Use the implicit function theorem to show that there exists $X : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^d$, $X(0) = 0$, such that $V(X(\mu), \mu) = 0$. Verify that, for μ small enough $\partial_x V(X(\mu), \mu)$ is hyperbolic. Hence A_K is open. For the density, if $V(0) = 0$ but $D_0 V$ is not hyperbolic, then show that one can find a matrix A such that, setting $V(x, \mu) := V(x) + \mu Ax$, holds $V(\cdot, \mu) \in A_K$ for all $\mu \neq 0$. This is better done in the coordinates in which $D_0 V$ is in normal Jordan form.)

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¹Note that, if K is compact, then finitely many such neighborhoods will cover K . On the other hand if, for example, $K = \mathbb{R}^d$, then countably many neighborhoods will do the job.

²The norm being $|V|_\infty + |DV|_\infty$.

Problem 3. Show that typically for each $\bar{x} \in \mathbb{R}^d$ one can find a neighborhood $U \ni \bar{x}$ in which $V(x) \neq 0$ with, at most, the exception of $\bar{x}$. (Hint: Assume that $V \in A_K$. If $V(\bar{x}) \neq 0$, then the existence of U such that $V(x) \neq 0$ for all $x \in U$ follows just by the continuity of V . If $V(\bar{x}) = 0$, then the other zeroes must satisfy $D_{\bar{x}}V(x - \bar{x}) + F(x) = 0$ where $\|F(x)\| = o(\|x - \bar{x}\|)$. But since $V \in A_K$, there exists $D_{\bar{x}}V^{-1}$ and this shows (e.g. use the contracting mapping theorem) that there exists a neighborhood in which the above equation has no solutions other than $\bar{x}$.)

Definition 2. We say that two vector fields V, W are equivalent in the open set U , if, for each $t > 0$, there exists a homeomorphism $F : U \rightarrow U$ such that, calling ϕ_t^V, ϕ_t^W the flows generated by the vector fields, holds $\phi_t^V \circ F = F \circ \phi_t^W$.

Finally, we say that two vector fields have the same local behaviour at $\bar{x}$ if it is possible to find $U \ni \bar{x}$ such that they are equivalent in U .

It is now clear that we understand already the typical local behaviour. In fact, either $V(\bar{x}) \neq 0$ and then the flow box Theorem tells us that the field has the same local behaviour than a constant vector field; or, if $V(\bar{x}) = 0$, then Grobmann-Hartman Theorem tells us that the field has the same local behaviour than its linear part.

2. GENERIC FAMILIES OF VECTOR FIELDS

Our next aim is to consider a situation in which the system has a control parameter. That is it is described by the equations

$$(2.1) \quad \dot{x} = V(x, \lambda)$$

where $x \in \mathbb{R}^d$ and $\lambda \in [-1, 1]$ is the parameter that, in principle, can be varied. Now by *local* understanding in a region K we mean that for each point $(\bar{x}, \bar{\lambda}) \in K \times [-1, 1]$ we can find a neighborhood of the form $U \times (\lambda - \varepsilon, \lambda + \varepsilon)$ in which are able to understand the behaviour of the solutions of (2.1). Typicality will instead be with respect to the space $\mathcal{C}^r(\mathbb{R}^d \times [-1, 1], \mathbb{R}^d)$, $r \geq 1$.

Let us now try to understand the local picture for typical families of vector fields. In analogy with the previous section, for compact $K \subset \mathbb{R}^d$, we can consider the set $\bar{A}_K := \{V \in \mathcal{C}^1 : V(x, \lambda) = 0 \text{ implies } \partial_x V(x, \lambda) \text{ hyperbolic } \forall (x, \lambda) \in K \times [-1, 1]\}$

Problem 4. Prove that if $V \in \bar{A}_K$, then for each $(\bar{x}, \bar{\lambda}) \in K \times [-1, 1]$ there exists an open set of the form $U \times (-\varepsilon + \bar{\lambda}, \varepsilon + \bar{\lambda}) =: U \times I$ such that either $V(x, \lambda) \neq 0$ or there exists $X \in \mathcal{C}^1(I, K)$ such that $V(X(\lambda), \lambda) = 0$ for each $\lambda \in I$ and there are no other zeroes in $U \times I$. (Hint: implicit function theorem.)

Clearly the above situations can be treated exactly as we did in the previous section and are therefore locally understandable. Unfortunately, the above does not exhaust all the possibilities.

Lemma 2.1. For each K with non empty interior $\bar{A}_K$ it is not generic.

Proof. Clearly $\bar{A}_K$ is open, the problem is the density. To see this let us consider, for example, the case $d = 1$, a compact set K with interior containing zero and the family

$$V(x, \lambda) = \lambda a + \lambda x + bx^2.$$

Now let us consider any $W \in \mathcal{C}^1(\mathbb{R} \times [-1, 1], \mathbb{R})$ and look at $\tilde{V}(x, \lambda, \mu) := V(x, \lambda) + \mu W(x, \lambda)$. The claim is that for each μ sufficiently small, then $\tilde{V}(x, \lambda, \mu) \notin \bar{A}_K$.

In fact, there exists $(x(\mu), \lambda(\mu)) \in K$ such that both $\tilde{V}(x(\mu), \lambda(\mu), \mu) = 0$ and $\partial_x \tilde{V}(x(\mu), \lambda(\mu), \mu) = 0$. To see this we define the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$F(x, \lambda, \mu) := \begin{pmatrix} \lambda a + \lambda x + bx^2 + \mu W(x, \lambda) \\ \lambda + 2bx + \mu \partial_x W(x, \lambda) \end{pmatrix} = \begin{pmatrix} \tilde{V} \\ \partial_x \tilde{V} \end{pmatrix},$$

clearly we are looking for $(x(\mu), \lambda(\mu))$ such that $F(x(\mu), \lambda(\mu)) = 0$. Since $F(0, 0, 0) = 0$ we can apply the implicit function theorem provided

$$\begin{pmatrix} 0 & a \\ 2b & 1 \end{pmatrix}$$

is invertible, that is if $ab \neq 0$. We have thus seen that the family has an open neighborhood disjoint from $\bar{A}_K$, hence the latter set cannot be dense. $\square$

Thus, to have a generic situation we need to consider a larger set.

A (not so obvious) possibility is given by the following. Before stating we need a bit of notation. For each $v \in \mathbb{R}^d$ let us define the $d \times d$ matrix valued function $H_v(x, \lambda) := \partial_x V(x, \lambda) + \partial_\lambda V(x, \lambda) \otimes v$ and the real valued function $D_v(x, \lambda) := \langle H_v(x, \lambda)^{-1} \partial_\lambda V(x, \lambda), v \rangle$.

$$B_K \{ V \in \mathcal{C}^2 : \forall (x, \lambda) \in K \times [-1, 1] \ V(x, \lambda) = 0 \text{ implies } \exists v \in \mathbb{R}^d : \exists H_v(x, \lambda)^{-1}; \\ \langle (\partial_x D_v, \partial_\lambda D_v), (H_v^{-1} \partial_x V v, \langle H_v^{-1} \partial_x V v, v \rangle) \rangle \neq 0 \}$$

Problem 5. Show that, for each compact set K , B_K is generic. (Hint: show first that the fact that $\partial_x V$ has a null space of dimension at most one is generic. If $\partial_x V$ is invertible, then the argument is the same as in Problem 2. If, instead, the null space is not empty, there exists $w \in \mathbb{R}^d$ such that $\langle w, \partial_x V \xi \rangle = 0$ for all ξ . On the other hand, if $\partial_x V(\bar{x}, \bar{\lambda}) + \partial_\lambda V(\bar{x}, \bar{\lambda}) \otimes v$ is not invertible there exist ξ such that $\partial_x V(\bar{x}, \bar{\lambda}) \xi + \partial_\lambda V(\bar{x}, \bar{\lambda}) \langle v, \xi \rangle = 0$. Multiplying for w follows $\langle w, \partial_\lambda V \rangle = 0$. Next, show that if $\langle w, \partial_\lambda V \rangle \neq 0$, then $V \in B_K$. If not then there exists $\tilde{w}$ such that $\langle \tilde{w}, \partial_x V(\bar{x}, \bar{\lambda}) \xi + \partial_\lambda V(\bar{x}, \bar{\lambda}) \langle v, \xi \rangle \rangle = 0$. Choosing $\xi = v$ follows $\langle w, \partial_\lambda V \rangle = 0$, but then $\langle \tilde{w}, \partial_x V \xi \rangle = 0$ for each ξ , which, by the simplicity of the zero eigenvalue, implies $\tilde{w}$ proportional to w , hence $\langle w, \partial_\lambda V \rangle = 0$ contrary to the assumption. Finally check that $\langle w, \partial_\lambda V \rangle = 0$ is generic. The genericity of the second condition follows again from the implicit function theorem.)

The point of B_K is that it allows to prove the analogous of Problem 3.

Lemma 2.2. If $V \in B_K$ and $V(\bar{x}, \bar{\lambda}) = 0$, then there exists $\varepsilon > 0$ and a neighborhood $U \ni \bar{x}$ such the vector field $V(x, \lambda)$ either has no zeroes or the set of zeroes consists of a smooth curve in $U \times (\bar{\lambda} - \varepsilon, \bar{\lambda} + \varepsilon)$. Moreover, the vector fields in such a curve belongs to A_K with, at most, one exception.

Proof. Now the approach based on a simple application of the implicit function theorem suggested in Problem 3 does not work if $\partial_x V(\bar{x}, \bar{\lambda})$ is not invertible, yet one can salvage such a situation in the following way. Suppose, without loss of generality, that $(\bar{x}, \bar{\lambda}) = (0, 0)$. Consider a vector $v \in \mathbb{R}^d$, $\|v\| = 1$, and the change of variables $(\lambda, x) = F_v(\xi, \mu)$ defined by

$$\begin{aligned} x &= \xi - \mu v \\ \lambda &= \langle \xi, v \rangle. \end{aligned}$$

It is easy to check that F^{-1} is defined by

$$\begin{aligned}\mu &= \lambda - \langle x, v \rangle \\ \xi &= \lambda v + x - \langle x, v \rangle v.\end{aligned}$$

Then

$$\det(DF^{-1}) = \det \left(\frac{1}{v_j} \middle| \frac{-v_i}{\delta_{ij} - v_i v_j} \right) = \det \left(\frac{1}{0} \middle| \frac{-v_i}{\delta_{ij}} \right) = 1.$$

Then define the field $\tilde{V} := V \circ F$. Since $F(0, 0) = 0$, $\tilde{V}(0, 0) = 0$. To apply the implicit function theorem in the new variable we need $\partial_\xi \tilde{V}$ to be invertible, but $\partial_\xi \tilde{V}(x, \lambda) = \partial_x V(x, \lambda) + \partial_\lambda V(x, \lambda) \otimes v$ so one can insure the invertibility, in a neighborhood of zero, since $V \in B_K$. We have then a $\mathcal{C}^1$ function $\xi(\mu)$ such that $\tilde{V}(\xi(\mu), \mu) = 0$, with $\xi'(\mu) = (\partial_x V + \partial_\lambda V \otimes v)^{-1} \partial_x V v$. This means that we have a smooth curve of singular points for V in a neighborhood of $(0, 0)$.

To conclude, we note that $\partial_x V(x(\mu), \lambda(\mu))$ has a non empty null space iff $\frac{d\lambda}{d\mu} = 0$. Indeed,

$$\frac{d\lambda}{d\mu} = \langle \xi'(\mu), v \rangle = 1 - \langle (\partial_x V + \partial_\lambda V \otimes v)^{-1} \partial_\lambda V, v \rangle =: \langle \zeta, v \rangle,$$

where, by definition, $\partial_x V \zeta + \partial_\lambda V \langle v, \zeta \rangle = \partial_\lambda V$, but if $\frac{d\lambda}{d\mu} = 0$, then $\langle \zeta, v \rangle = 1$, hence $\partial_x V \zeta = 0$. In conclusion, the relevant points where $\partial_x V$ is not invertible must lie on the manifold $D(x, \lambda) - 1 = 0$, while the curve of the zeroes of V moves in the direction $(\xi', \langle \xi', v \rangle)$, but then the condition $V \in B_K$ means exactly that the gradient of $D - 1$ and the latter vector are not perpendicular, in particular the gradient of D does not vanish. On the other hand, this latter fact implies that we can apply the implicit function theorem to see that the zeroes of $1 - D$ form a smooth manifold near $(0, 0)$ thus the gradient is perpendicular to such a manifold. Consequently, the curve $(x(\mu), \lambda(\mu))$ is transversal to the above manifold, hence it can have only an intersection (the point $(0, 0)$ indeed) in a sufficiently small neighborhood of zero. $\square$

Thus, typically, we have to worry only about families in which the vector field has, at most, one zero. In fact, due to the previous discussion, we need to consider only the case in which there is one zero and the derivative is not hyperbolic. Let Δ_d be the set of $d \times d$ matrices that are not nilpotent.

$$(2.2) \quad D_K := \{V \in B_K : V(x, \lambda) = 0 \text{ implies } \partial_x V(x, \lambda) \in \Delta\}.$$

Problem 6. Show that, for each $K \subset \mathbb{R}^2$ and $d \leq 2$, D_K is generic.

In conclusion, in one dimension we must analyze the case $V(x, \lambda) = \partial_x V(x, \lambda) = 0$, while in $d = 2$, to be able to have a complete local understanding, we need to analyze the cases in which $V(x, \lambda) = 0$ but $\partial_x V(x, \lambda) \in \Delta_2$ is not hyperbolic.

Problem 7. Prove that if $A \in \Delta_2$ then only the following possibilities can occur

- A is hyperbolic
- A has a zero eigenvalue
- A has a Jordan block
- A has two purely imaginary conjugated eigenvalues.

(Hint: put the matrix in Jordan normal form and see what it means to be nilpotent).

3. ONE DIMENSION

In the one dimensional case let K be a compact set containing a neighborhood of zero and $V \in D_K \setminus \bar{A}_K$, suppose, without loss of generality, that $V(0,0) = 0$. Then it must be

$$(3.1) \quad V(x, \lambda) = \lambda a + \lambda b x + c x^2 + g(x, \lambda),$$

where $|g(x, \lambda)| \leq C(x^2 + \lambda^2)$.

Problem 8. *Prove that the condition $c \neq 0$ is generic in $\mathcal{C}^2$.*

Then $V(x, \lambda)$ has no solutions if $ac > 0$, while for $ac < 0$ there are the two solutions $x = \pm \sqrt{\frac{\lambda a}{c}} + \mathcal{O}(\lambda)$. We have therefore the generic picture: either two points collide and kill each other or there is a creation of two zeroes of the vector field.

Problem 9. *Prove that the two equilibrium points of the vector field are one attractive and the other repulsive.*

The above scenario is called a *saddle-node* bifurcation.

A natural question is if there exists a simpler standard form of the above bifurcation. Indeed we can try to kill some of the terms in 3.1 by a change of variable.

Problem 10. *Show that with a change of variables of the type $z = \alpha + x + \beta x^2$, $\mu = \rho \lambda$ one can change the vector field (4.1) to the form $\bar{V}(z, \mu) = \mu + b z^2 + \mathcal{O}(z^2 + \mu^2)$.*

The above is the *normal form* of the saddle node bifurcation. This type of reduction can be made for each bifurcation and give rise to the large field of normal form theory which, unfortunately, goes beyond the scopes of the present notes.

4. TWO DIMENSIONS: A ZERO EIGENVALUE

In this case the vector field must have the form (possibly after a linear change of variable to put $\partial V_x(0,0)$ in normal form)

$$(4.1) \quad V(x, \lambda) = \bar{a}(\lambda) + A(\lambda)x + G(x, \lambda)$$

where $a(0) = 0$, $\|G(x, \lambda)\| \leq c\|x\|^2$ and

$$A(0) = \begin{pmatrix} 0 & 0 \\ 0 & \nu \end{pmatrix}$$

At this point it is easy to show that the scenario is exactly the same than in the one dimensional case. We leave the details to the reader.

5. TWO DIMENSIONS: A JORDAN BLOCK

In this case $A(0)$ is invertible.

Problem 11. *Show that, if $A(0)$ is invertible, then one can make a change of coordinates so that $\bar{a}(0) = 0$ in (4.1).*

We leave to the reader the task to understand the (not so exciting) scenario in this case.

6. TWO DIMENSIONS: EIGENVALUES WITH ZERO REAL PART

In this case we have the well known *Hopf* bifurcation. See related note.

7. THE HAMILTONIAN CASE

It is important to note that non generic situations may appear due to symmetries or other type of constraints. To give an example of such a situation let us consider an Hamiltonian vector field, that is a vector field of the type $V(x, p) = (\partial_p H, -\partial_x H)$ for some function $H(x, p)$. In this case

$$DV = \begin{pmatrix} \partial_{xp} H & \partial_{pp} H \\ -\partial_{xx} H & -\partial_{xp} H \end{pmatrix}.$$

Note that the trace of DV is always zero, hence if one eigenvalue is null also the other must be. It is thus clear that in this case the situation with two complex conjugate eigenvalues is generic for a vector field while two, not one, zero eigenvalues is generic for a vector field family. We must thus consider, for example, a family of the type

$$V(x, \lambda) = (p, -\lambda x - x^2)$$

Problem 12. *Show that in the above family we have the collision of two fixed point (a center and a saddle) that collide and exchange type.*

Remark 7.1. *Among other things, in this case we have a new situation even for the local understanding of vector fields due to the possibility of two complex conjugate eigenvalues a center. It is not obvious what is the local picture under perturbation for such a situation. For the two dimensional case, it implies that the Hamiltonian has a minimum and hence we have persistence of the center under perturbations. In higher dimensional situation this issue is the subject of the so called KAM theory. See the note on Siegel Theorem to have a glimpse in such a theory.*

Problem 13. *Consider the case of an Hamiltonian with a degenerate minimum, e.g. $H(x, p, \lambda) = \frac{1}{2}p^2 + x^4 + \lambda x$. What can you say?*

Remark 7.2. *The reader wishing to get a bit deeper in the bifurcation theory may look at [1, Chapter 6].*

REFERENCES

- [1] V.I. Arnold, *Geometric Methods in the theory of Ordinary Differential Equations*, Springer-Verlang, New York (1988).

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